

1. EXECUTIVE SUMMARY

This report documents all software engineering, cloud infrastructure, tax engine calculation, and workflow automation tasks completed for the **B2B Wholesale Wine Distribution Platform**.

The system was built as a multi-portal SaaS application featuring role-based access control, complex Belgian commercial tax calculations (HT Subtotal, 21% VAT, and €1.35 Vidange bottle deposits), serverless A4 delivery note PDF generation via headless Puppeteer in Firebase Cloud Functions, dynamic multi-language support (EN, FR, NL, UR), and real-time depot order dispatch management.

Key Project Metric: Total estimated development effort is **150 Hours** across 7 core software engineering streams. Commercial market valuation for an enterprise MVP of this scope is estimated between **€11,250 and €16,500**.

2. TASK & DELIVERABLES BREAKDOWN

Phase 1: Cloud Infrastructure & Database Architecture

- Firebase Setup & Config:** Initialized hosting environment, custom CORS header policies (cors.json), and deployment rules (firebase.json).
- NoSQL Firestore Schema:** Designed normalized schemas for wines (SKUs, HT prices, bottle deposit rates), orders (tax breakdown, delivery address, PDF URL), and users (RBAC rules).
- Cloud Storage Architecture:** Configured secure storage buckets for product images (wine-images/{sku}.jpg) and binary delivery note PDFs (invoices/{orderId}.pdf).

Phase 2: Admin Operations & Product Portal (/admin.html)

- Product CRUD Management:** Form interface supporting SKU creation, bottle & case HT price editing, 21% VAT, deposit per case (€1.35/bottle), and stock toggles (stockAvailable).
- Zero-Latency Image Upload Pipeline:** Local FileReader preview paired with automated Firebase Storage uploading and Data URL fallback for zero CORS friction.
- User & Store Manager:** Dynamic client shop store management (Food City Bruxelles, Wine Boutique Gent, Le Caveau Liège) and role assignments.
- Inventory & Barcode Utilities:** Automatic low-stock warnings (<10 cases) and integrated hardware/camera barcode scanner for fast inventory audits.
- Sales Analytics Dashboard:** Real-time reporting on total revenue (€), total cases sold, order count, and top SKUs.

Phase 3: Field Evaluator Ordering Web App (/index.html)

- Multi-Language i18n Engine:** Full support for English, French, Dutch, and Urdu with dynamic LTR/RTL layout switching.
- Client Shop Selector Guard:** Enforces client selection prior to ordering, populating store VAT numbers and delivery addresses.
- Belgian Tax & Vidange Calculation Engine:** Real-time cart calculations showing HT Subtotal, 21% Belgian VAT, €1.35 Vidange bottle deposits, and TTC Grand Total.

Phase 4: Serverless Headless PDF Invoice Engine (functions/index.js)

- Node 20 Firebase Cloud Function:** Event-driven function (generateNoteDenvoi) deployed in europe-west9 with 1GiB memory allocation.

- **Headless Puppeteer PDF Compiler:** Utilizes @sparticuz/chromium to render A4 Belgian "NOTE D'ENVOI" delivery note PDFs automatically upon order placement, uploading binary PDFs to Firebase Storage and updating order records with signed download links.

Phase 5: Warehouse Depot Dispatch Queue (/depot.html)

- **Real-time onSnapshot Order Stream:** Live order queue updating instantly as orders are placed.
- **Dispatch Lifecycle:** Workflow state tracking (submitted → ready_for_dispatch → dispatched) with 1-click PDF download links for packing slips.

Phase 6 & 7: Security Guards, QA & Technical Documentation

- **RBAC & Security Rules:** Implemented auth-guard.js route protections and Firestore/Storage security rules enforcing strict role boundaries.
- **Developer Documentation:** Comprehensive technical docs written in README.md, PROJECT_SPEC.md, and IMPLEMENTATION_PLAN.md.

3. ESTIMATED DEVELOPMENT EFFORT BREAKDOWN

Module / Work Stream	Key Scope Items	Hours
1. Architecture & Cloud Infrastructure	Firebase Hosting, Firestore DB schema, Storage CORS setup, deployment rules	16 hrs
2. Admin Management Portal	Product CRUD, pricing HT, 21% VAT, Vidanges, image fallback, low stock alerts, barcode scanner, sales BI	32 hrs
3. Field Evaluator Mobile Web App	Client shop selector, Belgian tax & deposit engine, multi-language i18n (EN/FR/NL/UR) + RTL	30 hrs
4. Headless PDF Cloud Function	Node 20 Cloud Function, Puppeteer @sparticuz/chromium, A4 Note d'Envoi compilation & storage link	24 hrs
5. Depot Dispatch Queue	Real-time onSnapshot stream, order status lifecycle management, PDF print action	20 hrs
6. Security & RBAC Guards	auth-guard.js, Firestore rules, Storage security rules, user role seeding	14 hrs
7. QA, Optimization & Documentation	CORS resolution, cross-browser testing, README.md, PROJECT_SPEC.md, implementation plan	14 hrs
TOTAL ESTIMATED DEVELOPMENT EFFORT		150 Hours

4. INDUSTRY STANDARD CHARGING BENCHMARKS

Billing Model / Tier	Rate Structure	Total Valuation (~150 hrs)	Market Benchmark Context
Senior Freelancer (EU / US)	€75 – €110 / hr (\$80 – \$120/hr)	€11,250 – €16,500	Standard rate for experienced full-stack engineers in Western Europe.
Specialized Software Agency	€120 – €165 / hr	€18,000 – €24,750	Standard rate charged by Belgian/European digital agencies.
Fixed-Price MVP Package	Lump Sum Fee	€12,500 – €15,000	Recommended package pricing for custom B2B SaaS MVP scope.

5. ITEMIZED CLIENT INVOICE DRAFT

Item	Description	Hours	Rate (€)	Amount (€)
01	Cloud Infrastructure & Database Schema Design	16	€85.00	€1,360.00
02	Admin Operations & Product Management Portal	32	€85.00	€2,720.00
03	Field Evaluator Ordering App & Tax/Vidange Engine	30	€85.00	€2,550.00
04	Serverless Headless PDF Invoice Cloud Function	24	€85.00	€2,040.00

Item	Description	Hours	Rate (€)	Amount (€)
05	Depot Dispatch Queue & Real-Time Stream	20	€85.00	€1,700.00
06	Security, RBAC & Authentication Guards	14	€85.00	€1,190.00
07	QA, Optimization & Technical Documentation	14	€85.00	€1,190.00
SUBTOTAL (150 Hours @ €85/hr)				€12,750.00
VAT (21% Belgian VAT, if applicable)				€2,677.50
TOTAL AMOUNT DUE				€15,427.50